

Professional Summary

Results-driven Python Backend and Cloud Engineer with 4+ years of experience designing, developing, and deploying scalable, secure, and cloud-native serverless applications. Proven expertise in Python AWS Lambda, EKS microservices, event-driven architectures, and Infrastructure as Code using AWS CDK, while leveraging AI-assisted development tools to accelerate delivery.

- Designed and developed scalable **RESTful APIs and microservices using Python (Flask)** with modular architecture and clean coding standards.
- Built responsive, user-centric web applications using **React, Angular, JavaScript (ES6+), HTML5, and CSS3**.
- Architected microservices-based systems with **API versioning, Swagger/OpenAPI documentation**, and service orchestration best practices.
- Implemented secure authentication and authorization mechanisms using **OAuth2, JWT, and OWASP security standards**.
- Designed and optimized relational and NoSQL databases using **PostgreSQL, MySQL, MongoDB**, with schema migrations managed via **Alembic**.
- Improved system performance through **Redis caching, query indexing, and backend performance tuning techniques**.
- Developed event-driven and asynchronous systems leveraging **Apache Kafka** for real-time data streaming.
- Deployed and managed cloud-native applications on **AWS (EC2, S3, RDS, Lambda, ECS)** ensuring scalability and high availability.
- Implemented **Infrastructure as Code (Terraform)** for automated environment provisioning and consistency.
- Applied containerization and DevOps best practices using **Docker, Kubernetes (fundamentals), and CI/CD pipelines (GitHub Actions, Jenkins, GitLab CI)**.
- Established monitoring and observability solutions using **Prometheus, Grafana, ELK Stack, and AWS CloudWatch**.
- Actively collaborated in **Agile/Scrum environments**, implementing automated testing using **PyTest, Unittest**, and API validation with **Postman**.

TECHNICAL STACK

Domain	Technologies
Backend	Python (FastAPI, Flask), RESTful APIs, Microservices, Event-Driven Architecture, AsyncIO, OOP
Frontend	React, TypeScript, GraphQL, JS (ES6+), HTML5, CSS3, REST Integration
DB/Cache	Amazon DynamoDB, PostgreSQL, MySQL, Redis, Amazon S3, Alembic
Cloud/DevOps	AWS (Lambda, Step Functions, EventBridge, EKS, API Gateway, VPC, IAM), AWS CDK (TypeScript), Docker, Kubernetes, Helm Charts, CI/CD
Messaging	AWS SQS, AWS SNS, Apache Kafka, Event-Driven Architecture
Testing	Pytest, Unittest, Playwright, Postman
Tools	Git, Claude Code (AI Assistant), GitHub Actions, Jenkins, Swagger
Security/Obs	AWS IAM, VPC Security, OAuth2, JWT, AWS CloudWatch, Prometheus, Grafana

EXPERIENCE:

Molina Healthcare, Inc.
Software Engineer (Python Developer)

Long Beach, CA
February 2024 – Present

Responsibilities:

- Architected and developed scalable healthcare applications using **Python (Flask)** in a microservices architecture, integrating with **Java Spring Boot services**, ensuring **HIPAA-compliant secure data handling**.
- Designed and implemented high-performance **RESTful APIs** using Flask, leveraging **PostgreSQL**, optimized indexing, and **Alembic-based schema versioning** for transactional claims and eligibility systems.
- Built modular backend components following **SOLID principles, OOP, and clean architecture patterns**, supporting 5M+ Medicaid users in a high-availability production environment.
- Implemented **Infrastructure as Code (IaC)** using **Terraform**, provisioning AWS resources (EC2, Lambda, IAM, VPC) for scalable, fault-tolerant deployments.
- Developed event-driven data pipelines using **Python automation scripts**, integrating **AWS S3, Lambda, EventBridge**, and asynchronous workflows for real-time ingestion and transformation.
- Integrated centralized logging and observability using **ELK Stack (Elasticsearch, Logstash, Kibana)** and **AWS CloudWatch**, improving MTTR by 40% in production systems.
- Engineered secure authentication mechanisms using **OAuth2, JWT tokens, RBAC**, and implemented API-level security aligned with OWASP standards.
- Led backend development for a conversational AI chatbot service, designing scalable API endpoints, managing session state, and optimizing request-response latency in distributed environments.
- Conducted backend architecture reviews, identifying scalability bottlenecks, improving API throughput, and recommending **horizontal scaling strategies** in AWS.
- Mentored engineers on **Python best practices, REST API standards, microservices communication patterns, CI/CD workflows, and secure cloud deployments**.

Skillset: **Python (Flask, FastAPI), REST APIs, Microservices, OOP, AsyncIO, PostgreSQL, MySQL, MongoDB, Redis, Alembic, Kafka, AWS (EC2, S3, RDS, Lambda, ECS), Docker, Kubernetes, Terraform, CI/CD, React, JavaScript (ES6+), OAuth2, JWT, ELK, Prometheus, Git.**

Responsibilities:

- Engineered RESTful backend APIs using **Python and Flask**, enabling real-time industrial telemetry tracking and rule-based alert orchestration.
- Implemented modular service architecture with **Blueprints, WSGI, and middleware layers** to support scalable manufacturing dashboards.
- Managed database schema migrations using **Alembic and SQLAlchemy ORM**, ensuring version-controlled deployments and backward compatibility.
- Designed optimized relational data models (PostgreSQL/MySQL) with indexing, constraints, and query performance tuning.
- Integrated AWS services (**S3, EC2, CloudWatch, IAM**) to build secure, event-driven data ingestion and monitoring pipelines.
- Developed asynchronous background jobs using **Celery and Redis**, automating scheduled data refresh and alert processing tasks.
- Created cron-based automation scripts and self-healing monitoring utilities using **Python multiprocessing and logging frameworks**.
- Refactored monolithic Java components into lightweight **Flask-based microservices**, improving API latency and maintainability.
- Built event-driven notification engine leveraging **Python decorators, custom rule evaluators, and JSON-based configuration parsing**.
- Implemented unit and integration testing using **PyTest, unittest, and mock libraries**, increasing backend test coverage.
- Optimized API performance with caching strategies using **Redis and Flask-Caching**, reducing response time for high-volume endpoints.
- Collaborated in Agile workflows (Sprint Planning, Code Reviews, CI/CD pipelines) integrating **Git, Docker, and AWS deployment strategies**.

Skillset: Python, Flask, SQLAlchemy, Alembic, REST APIs, Celery, Redis, AWS (S3, EC2, CloudWatch, IAM), PostgreSQL, Docker, Git, PyTest, CI/CD, Microservices Architecture, Cron Automation

Responsibilities:

- Developed foundational backend components for engineering test tools using **Python, Flask, and MySQL**, following Agile and SDLC best practices.
- Built and enhanced secure **RESTful APIs in Python (Flask)** with token-based authentication and role-based access control for Angular-based dashboards.
- Implemented backend business logic using **Python OOP principles, modular architecture, and reusable service layers**.
- Designed and executed unit and integration tests using **Python unittest and PyTest frameworks**, improving pre-release defect detection.
- Wrote optimized **SQL queries and stored procedures** for high-volume backend data extraction, validation, and reporting accuracy.
- Automated backend validation scripts using **Python scripting and logging modules** to streamline engineering test data processing.
- Created API documentation using **Swagger/OpenAPI specifications**, improving developer onboarding and cross-team integration.
- Participated in backend prototyping using Python-based microservices to validate system scalability and technical feasibility.
- Developed system health monitoring and reporting automation using **Python cron jobs and scheduled scripts**.
- Assisted in debugging and performance tuning of Flask applications using profiling and logging tools.
- Collaborated with frontend teams to structure clean JSON responses and ensure seamless API consumption.
- Conducted technical walkthroughs and knowledge transfer sessions, explaining Python backend workflows and API integrations to QA and support teams.

Skillset: Python, Flask, REST APIs, MySQL, SQL Optimization, PyTest, unittest, Swagger/OpenAPI, OOP, Microservices, Cron Jobs, JSON, Agile, Git

PROJECTS:

E-Commerce Order Management Platform | Java, Spring Boot, Go, Python, Flask, PostgreSQL, Redis, Docker, AWS (ECS, S3, RDS), Apache Kafka, Terraform, Prometheus, Grafana

- Developed cloud-native backend features for transactional services using Python (Flask) and AWS Lambda, deploying containerized microservices via Amazon EKS and orchestrating Infrastructure as Code via AWS CDK (TypeScript) following serverless best practices.
- Architected highly resilient asynchronous workflows using AWS Step Functions, EventBridge, and SQS/SNS to enable real-time message routing, payment confirmations, and decoupled system updates at scale.
- Designed and optimized scalable data models using Amazon DynamoDB and PostgreSQL schemas, leveraging indexed read-heavy queries and S3 storage integrations to reduce checkout latency by 45% during peak traffic.
- Instrumented system observability and performance monitoring using AWS CloudWatch, Prometheus, and Grafana, ensuring 99.9% uptime and streamlined debugging across core transactional and API Gateway routes.

Community Connect – Blockchain App | Solidity, React, Web3.js, Ethereum, IPFS, Truffle

- Engineered robust AWS Lambda backend handlers and REST APIs in Python to manage user authentication, API Gateway endpoints, and AWS IAM role-based access controls.
- Built a responsive and high-performance frontend using React and TypeScript, integrating GraphQL APIs to securely handle multi-step workflows, UI state transitions, and user interactions.
- Automated continuous integration and delivery pipelines (CI/CD) utilizing Helm Charts and Docker for seamless microservice deployments to Amazon EKS environments.
- Conducted comprehensive end-to-end backend testing using Pytest and automated frontend integration tests using Playwright, leveraging AI coding assistants to speed up test execution and accuracy.

EDUCATION:

